From Linear Models to Transformers

Same pixels, five inductive biases: image classification on Fashion-MNIST

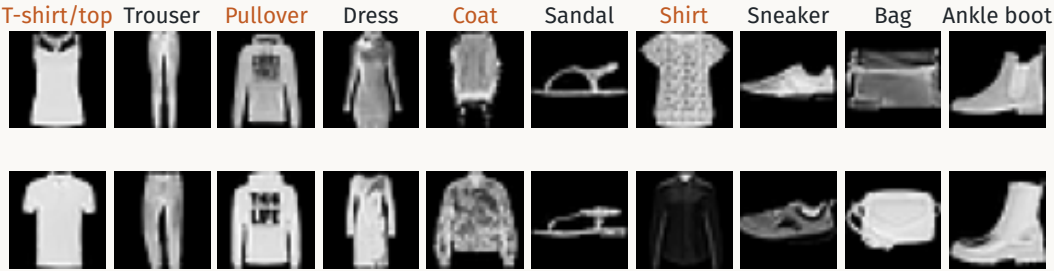
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CO3133 Deep Learning and Its Applications Assignment 1 Semester 261 Instructor: Lê Thành Sách

Outline

1	Problem and data	10 classes, 4 look alike
2	Method	$f = h \circ \varphi$
3	Setup	one protocol, 3 seeds
4	Results	macro-F1 vs. cost
5	Error analysis	Shirt
6	Conclusion	bias over capacity

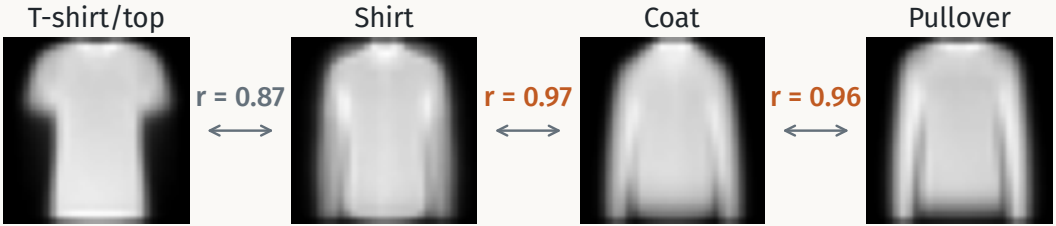
Task



54,000 train / 6,000 val / 10,000 test 5,400 / 600 / 1,000 per class 0 duplicates across splits

> Same pixels, same protocol: how much of the result does the inductive bias decide?

Where it is hard



Pearson correlation between class-mean training images

Shirt, Coat and Pullover average to almost the same image: $r \geq 0.95$. Errors will live here.

Same pixels, four layouts

Linear, MLP



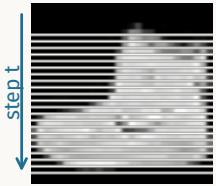
flat vector, 784

CNN



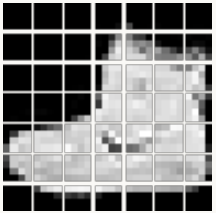
grid, 1×28×28

LSTM, GRU



28 rows × 28

Transformer



49 patches × 16

φ: fixed, parameter-free rearrangement of the 784 pixels h: learned network, outputs 10 logits

> $f = h \circ \varphi$, and every φ is a bijection: any gap between models comes from h.

Five families

Model	Input	Architecture	Params
Linear	vector	one affine layer	7,850
MLP	vector	512, 256 hidden; BatchNorm, ReLU, dropout 0.3	536,586
CNN	grid	3 blocks (conv3×3, BN, ReLU)×2 + pool; 32/64/128; GAP	288,170
LSTM	rows	2 layers, hidden 128, last state → linear head	214,282
GRU	rows	identical to the LSTM except the cell	161,034
Transformer	patches	d=96, [CLS] token, learned pos. enc., 4 blocks × 4 heads	306,826

All implemented from scratch, no pretrained weights. Sizes are not matched: 7.9k to 537k.

Protocol

Factor	Setting
Split	stratified 90 / 10 of official train, seed 42; test set used once
Optimiser	AdamW, lr 10^{-3}, batch 128, gradient clipping 1.0
Schedule	1 epoch warm-up, cosine decay; 40 epochs, early-stop patience 10
Checkpoint	best validation macro-F1
Seeds	42, 123, 2026 $\rightarrow$ mean $\pm$ std

Identical for every model, fixed in advance, not tuned per family RTX 3060 51 runs, about 1.5 GPU-hours

> A beats B only if Δmacro-F1 $>$ the larger seed std **and** McNemar rejects at $p < 0.05$.

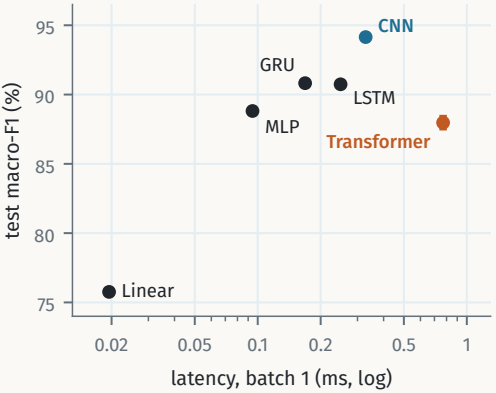
Fashion-MNIST

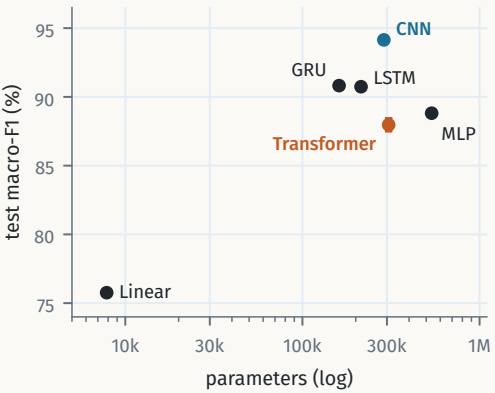
Model	Params	Macro-F1 (%)	Train (s)	Latency (ms)
CNN	288,170	**94.15 ± 0.14**	94	0.33
GRU	161,034	90.82 ± 0.21	47	0.17
LSTM	214,282	90.74 ± 0.12	48	0.25
MLP	536,586	88.81 ± 0.12	39	0.09
Transformer	306,826	**87.97 ± 0.47**	131	0.77
Linear	7,850	75.76 ± 0.18	32	0.02

Test set, mean ± std over 3 seeds latency at batch 1, FP32 most runs still improving at epoch 40: lower bounds

CNN first. Every neighbouring pair differs conclusively, except LSTM vs. GRU ($p = 0.55$).

Quality vs. cost

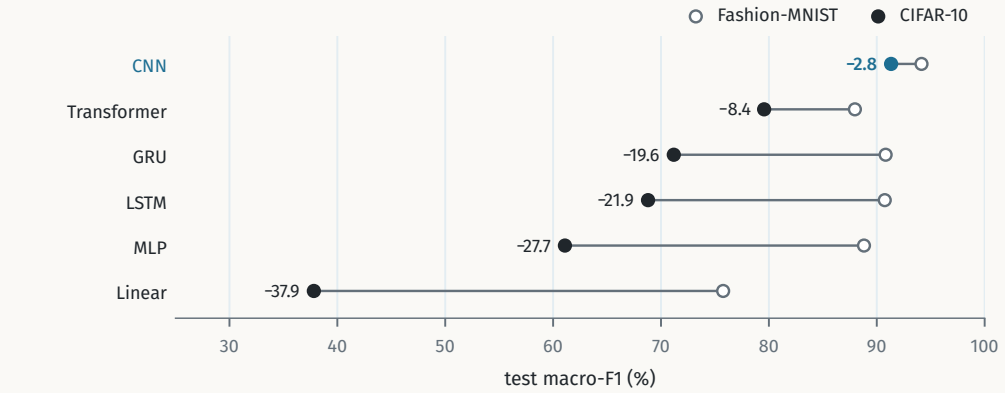




Fashion-MNIST, mean ± std over 3 seeds, log x-axes

GRU: 96.5% of the CNN's macro-F1 with 56% of its parameters at half the latency.

Extension: CIFAR-10



CIFAR-10 test macro-F1, mean over 3 seeds 60 epochs, patience 15 labels: drop in points

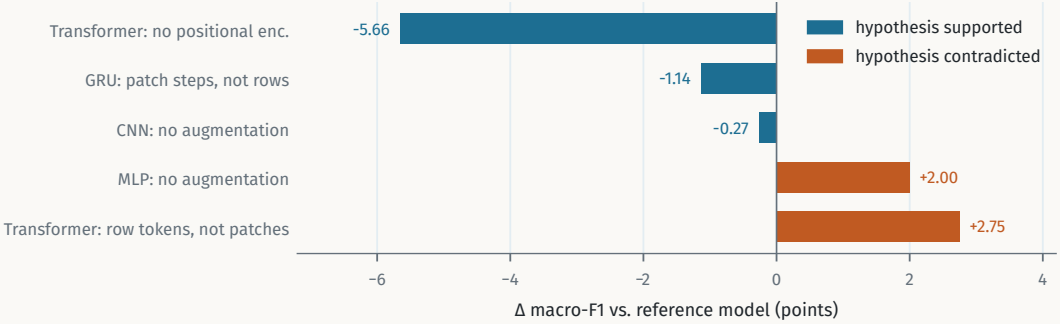
Colour, cluttered scenes: the CNN loses 2.8 points, the MLP 27.7. The CNN–MLP gap grows 5.3 → 30.2.

Extension: LSTM vs. GRU

	Fashion-MNIST		CIFAR-10	
	LSTM	GRU	LSTM	GRU
Parameters	214,282	161,034	891,402	669,194
Macro-F1 (%)	90.74 ± 0.12	90.82 ± 0.21	68.80 ± 0.29	$\mathbf{71.19 \pm 0.17}$
Latency (ms)	0.25	0.17	0.50	0.45
McNemar p	0.55		5.5×10^{-9}	

Fashion-MNIST: no conclusive difference, GRU cheaper everywhere. CIFAR-10: GRU also better.

Ablations

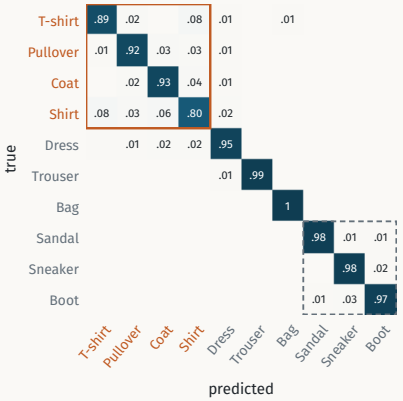


Fashion-MNIST, 3 seeds each, one factor changed vs. the main model hypotheses written before the runs

Positional encoding is worth 5.7 points. Two hypotheses failed: row tokens help, augmentation hurts the MLP.

Where the CNN fails

- **66.8%** inside the upper-body block
- 11.3% inside footwear
- Shirt ↔ T-shirt alone: 25.8%
- Shirt: lowest F1 for every model



CNN, seed 42: 594 errors on 10,000 test images, row-normalised confusion

> Two thirds of the CNN's errors stay inside the upper-body group.

Confident and wrong



CNN, seed 42: the upper-body cases among its 25 most confident errors, all at confidence ≥ 0.98

144 errors (24.2%) above 0.9 confidence. Plain, collarless garments: the cue is a few pixels.

Demo: one run, end to end

Command (in `scripts/`)	Produces
`train.py --config configs/fashion_mnist/cnn.yaml --seed 42`	checkpoint, test metrics
`evaluate.py --run-dir outputs/runs/fashion_mnist/cnn/seed42`	confusion, error galleries
`compare.py --dataset fashion_mnist`	tables, McNemar, figures

> Every number in this talk traces back to a config, a split fingerprint, a checkpoint hash and a commit.

Conclusion

Findings

- CNN first on both datasets, more so on CIFAR-10
- No single winner on cost: GRU for latency
- Transformer: tokens matter, +2.75 with rows

Limits

- one untuned protocol, unmatched sizes
- 40-epoch budget: values are lower bounds
- 3 seeds; McNemar on seed 42 only

> Matching inductive bias beats raw capacity: the CNN beats an MLP with 1.9× its parameters.

Report, code and slides: `pbt245.github.io/CO3133-DLIA`